

A Probabilistic Shared Control Framework for $SE(3)$ Grasping

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Abstract—Teleoperating a high-degree-of-freedom robot through a low-bandwidth interface is demanding, particularly when a task requires full- $SE(3)$ control. We present Robot Trajectron V3 (RT-V3), a trajectory-level probabilistic shared-control framework for multi-object, multi-affordance grasping. Instead of reducing intent to a discrete target, RT-V3 represents it as a multimodal distribution over future end-effector trajectories. A transformer-based conditional generative prior jointly reasons over recent robot dynamics, point-cloud geometry, and dense candidate grasp poses. At run time, Bayesian inference fuses this prior with noisy user commands to update the intent distribution and generate assistive actions. An asynchronous mode continues inferred coarse motion during input pauses while preserving fine-grained user authority near the target. Trained on 568,277 simulated trajectories, RT-V3 attains a 90.4% closed-loop planning success rate and achieves the highest success rate across four simulated input impairments. In a study with 21 novice participants, RT-V3 reaches 86.9% grasp success and requires fewer joystick inputs and shorter completion times than direct control, while also reducing reported mental demand and effort. These results support trajectory-level inference as a practical approach to shared $SE(3)$ manipulation.

I. INTRODUCTION

Controlling a robot arm with many degrees of freedom through a joystick, head controller, or neural interface requires the user to map a small number of noisy input channels onto translational and rotational motion. Repeated mode switching and corrective commands make precise grasping especially demanding. Shared control can reduce this burden by combining user commands with autonomous assistance [1]. However, grasping in realistic scenes introduces two coupled ambiguities: which object the user intends to grasp and which of its many valid $SE(3)$ grasp poses they prefer.

Most shared-autonomy systems infer a discrete goal and then invoke a goal-conditioned controller [2], [3]. This formulation works well for small goal sets, but dense grasp candidates produce a large discrete goal space. Moreover, selecting a grasp does not capture how the user intends to approach it. Robot Trajectron V2 instead represents intent as a distribution over future trajectories and performs Bayesian action fusion [4]; however, it is restricted to planar translation and structured occupancy maps.

RT-V3 extends trajectory-level probabilistic shared control to cluttered, full-pose grasping. It (i) learns a multimodal behavioral prior from robot motion, point-cloud geometry, and candidate grasps; (ii) exposes a translation-conditioned rotational density that can be fused with interface uncertainty; and (iii) enables intermittent input through asynchronous

execution. We evaluate these capabilities through trajectory-prediction and planning ablations, large-scale tests with four simulated input impairments, and a real-world study with 21 novice participants. Additional technical details and experimental videos are available at https://mousecpn.github.io/RTV3_page/.

II. RT-V3 IN BRIEF

Figure 1 summarizes the probabilistic shared-control formulation of RT-V3. Let s_t denote the state at time t , including robot dynamics and scene context, and let a_t denote the velocity action. Taking $t = 0$ as the current time step, we define an intended trajectory as $\zeta = (s_{-T+1:H}, a_{-T+1:H-1})$, where $(s_{-T+1:0}, a_{-T+1:-1})$ is the observed past and $(s_{1:H}, a_{0:H-1})$ is the future. We learn the behavioral prior from successful demonstrations by maximizing

$$\max_{\zeta} \mathbb{E}_{\zeta} [\log p(s_{1:H}, a_{0:H-1} \mid s_{-T+1:0})]. \quad (1)$$

To factorize Eq. 1, we introduce a latent state h_t that summarizes the state history $s_{\leq t}$ and represent the future distribution as $p(h_{0:H}, a_{0:H-1} \mid s_{-T+1:0})$. Assuming first-order Markov dynamics, the distribution factorizes as

$$\begin{aligned} p(h_{0:H}, a_{0:H-1} \mid s_{-T+1:0}) \\ = p(h_0 \mid s_{-T+1:0}) \prod_{t=0}^{H-1} p(h_{t+1} \mid h_t, a_t) p(a_t \mid h_t). \end{aligned} \quad (2)$$

Here, the two factors in the product are the state-transition and action distributions.

To capture the pronounced multimodality of user intent, we adopt a conditional variational autoencoder (CVAE) formulation [5], [6] with a categorical latent variable z that indexes a low-dimensional set of maneuver modes:

$$\begin{aligned} p(h_{0:H}, a_{0:H-1} \mid s_{-T+1:0}) \\ = p(h_0 \mid s_{-T+1:0}) \sum_z p(z \mid h_0) \\ \prod_{t=0}^{H-1} p(h_{t+1} \mid h_t, a_t, z) p(a_t \mid h_t, z). \end{aligned} \quad (3)$$

Relative to Eq. 2, Eq. 3 introduces the maneuver-class prior $p(z \mid h_0)$ and conditions both the transition and action distributions on z [4].

Following standard CVAE training [7], [5], we minimize the following β -weighted negative evidence lower bound

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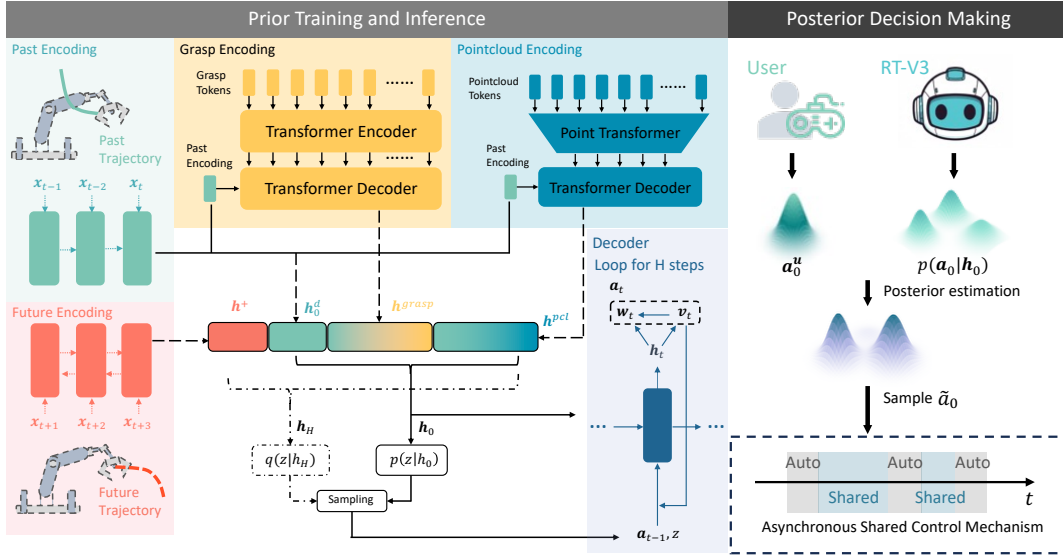


Fig. 1. Overview of RT-V3. A conditional generative prior encodes motion history, candidate grasps, and point-cloud geometry to predict a multimodal distribution over future $SE(3)$ trajectories. During teleoperation, each noisy user command updates the maneuver and current-action distributions. When input pauses, asynchronous control executes the most likely prior action until the robot enters the fine-adjustment region near an object.

(ELBO):

$$\begin{aligned} \mathcal{L}_{ELBO} = & -\mathbb{E}_{z \sim q(z|h_H)} [\log p(h_0 | s_{-T+1:0}) \\ & + \sum_{t=0}^{H-1} (\log p(h_{t+1} | h_t, a_t, z) + \log p(a_t | h_t, z)) \\ & + \beta D_{KL}(q(z | h_H) \| p(z | h_0))]. \end{aligned} \quad (4)$$

The training-time posterior $q(z | h_H)$ observes the complete demonstration and guides the deployable prior $p(z | h_0)$ through the Kullback–Leibler (KL) term. Optimizing Eq. 4 yields a multimodal behavioral prior over future trajectories.

At the current time step, RT-V3 observes a user command a_0^u received through an uncertain interface. Conditioning on this command yields the posterior

$$p(h_{0:H}, a_{0:H-1} | s_{-T+1:0}, a_0^u). \quad (5)$$

A maximum-a-posteriori estimate of Eq. 5 yields the intended action sequence $a_{0:H-1}$, of which RT-V3 executes a_0 . To make the update tractable, RT-V3 factorizes the twist density as $p(a_t | h_t, z) = p(w_t | h_t, z, v_t)p(v_t | h_t, z)$, models both factors as Gaussians, and fuses them with Gaussian interface uncertainty. The command also updates the maneuver weights through $p(z | h_0, a_0^u) \propto p(a_0^u | h_0, z)p(z | h_0)$.

Asynchronous control. When a command arrives, RT-V3 performs the posterior update above. During an input pause, it executes the most probable action under the learned prior, allowing the robot to continue its coarse approach without continuous user signaling. Prior-only execution is disabled within a predefined distance of an object surface so that fine grasp corrections remain closely coupled to the user.

III. EXPERIMENTS

A. Simulation Dataset and Prediction

We generated 22,265 tabletop scenes from 303 objects and sampled 833,324 feasible grasps. The scenes were split 9:1 into training and validation sets. For each feasible grasp, CuRobo [8] planned a motion from a random initial pose, which was then executed by a proportional–derivative (PD) controller. After discarding trajectories involving collisions, joint-limit violations, or failed grasps, the dataset contained 568,277 trajectories. We report average and final displacement errors (ADE/FDE) using a pose metric that combines translation with scaled Euler-angle error. Best-of-Mode (BoM) reports the error of the best-matching predicted maneuver, whereas Most Likely (ML) evaluates the highest-probability maneuver.

Scenes were created by randomly placing or dropping objects within a $30 \times 30 \times 30 \text{ cm}^3$ tabletop volume and acquiring depth observations from randomized viewpoints. The separate planning benchmark used 1,000 trials with fixed random seeds and two target grasps per scene. A trial succeeded if RT-V3 reached either target without contacting the table or an object; otherwise, it ended in collision or timeout. At each control step, RT-V3 selected the most likely maneuver and action. This benchmark therefore evaluates closed-loop robustness rather than Best-of-Mode forecasting.

Explicit grasp conditioning produces the largest reduction in prediction error (Table I). Adding point-cloud encoding further reduces the BoM errors, although the ML errors increase slightly, and raises planning success from 78.8% to 90.4%. The unfactorized 6-D variant yields the lowest open-loop errors but reaches only 3.4% success in closed-loop planning, consistent with compounding errors moving the policy outside its training distribution. The factorized RT-V3 reaches 90.4% success, slightly below the global

TABLE I

TRAJECTORY PREDICTION AND CLOSED-LOOP PLANNING. ADE/FDE DENOTE AVERAGE/FINAL DISPLACEMENT ERROR; BoM/ML DENOTE BEST-OF-MODE/MOST LIKELY; AND SR/CR/TOR DENOTE SUCCESS/COLLISION/TIMEOUT RATE. GREEN AND BLUE MARK THE BEST AND SECOND-BEST VALUES.

Method Components			Trajectory Prediction Metrics ($\downarrow$)				Planning Metrics		
Action Factorization	Grasp Enc.	PCL Enc.	ADE-BoM	FDE-BoM	ADE-ML	FDE-ML	SR ($\uparrow$)	CR ($\downarrow$)	TOR ($\downarrow$)
✓			42.71	83.79	46.40	92.89	-	-	-
✓	✓		26.95	38.12	30.36	44.17	78.8	10.3	10.9
✓	✓	✓	26.70	37.90	32.05	47.87	90.4	7.9	1.7
	✓	✓	21.05	34.55	22.50	37.65	3.4	68.7	27.9
CuRobo [8]			-	-	-	-	95.9	0	-
NeoSS [9]			-	-	-	-	83.5	5.4	11.1

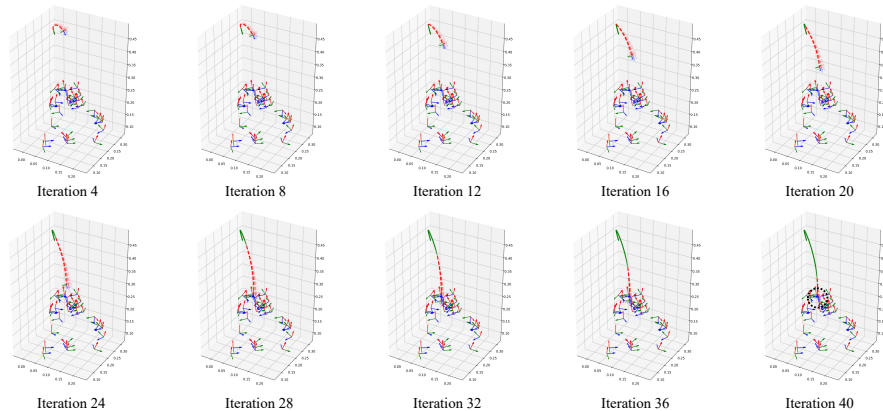


Fig. 2. RT-V3 trajectory predictions, shown every fourth iteration. Solid green curves show observed motion, dashed red curves show predicted futures, and coordinate frames denote terminal predictions and candidate grasps. Opacity encodes $p(z | h_0)$. The circle at iteration 40 highlights predictions that reach the intended grasp poses.

TABLE II

REAL-WORLD TASK OUTCOMES (%); UNMATCHED DENOTES FAILURES CAUSED BY A DISRUPTIVE SCENE CHANGE.

Method	Success $\uparrow$	Unmatched $\downarrow$	Timeout $\downarrow$
Direct	78.57	16.67	4.76
HO	74.70	18.07	7.23
RT-V3	86.90	9.53	3.57

planner CuRobo (95.9%) but above the reactive baseline NeoSS (83.5%), while retaining multimodal intent prediction.

B. Shared Control with Simulated Users

To evaluate assistance at scale, we used a common benchmark of 768 target grasps drawn from 50 held-out scenes, with 10–20 targets per scene. The same benchmark was evaluated under four simulated input impairments. Each user model first generated a goal-directed 6-D twist in the local frame. The impairment then added Gaussian noise (noisy), repeated the previous command with probability 0.8 (laggy), alternated between translation-only and rotation-only control (mode-switching), or retained only the degree of freedom (DoF) with the largest pose error (single-DoF). We compare direct teleoperation, Hindsight Optimization (HO), optional NeoSS collision avoidance, RT-V3, and an

unfactorized 6-D ablation. Table III reports success rate (SR), collision rate (CR), and mean iterations over successful trials.

RT-V3 achieves the highest SR under all four impairments, improving over Direct by 21.7, 5.7, 9.1, and 19.3 percentage points for noisy, laggy, mode-switching, and single-DoF input, respectively. NeoSS substantially reduces collisions, but Direct + NeoSS remains below RT-V3 in SR, indicating that collision avoidance alone does not recover user intent. HO’s lower SR is consistent with an observed failure mode in which a dense candidate set attracts the robot toward an incorrect grasp. Combining NeoSS with RT-V3 reduces CR to 0.3–5.6% while lowering SR by approximately two percentage points. The unfactorized model completes successful trials in fewer iterations but has lower SR and higher CR, supporting the closed-loop benefit of action factorization despite its lower open-loop errors.

C. Real-World User Study

Twenty-one novice participants (19 male and 2 female; ages 18–40) used an Xbox joystick to control a Franka Research 3 robot. The protocol assigned each participant three four-trial blocks, one each with Direct, HO, and RT-V3. Before each block, the controller was explained and participants completed practice trials. Both method order and the assignment of scene groups to methods were randomized. Each

TABLE III

SIMULATED SHARED CONTROL UNDER FOUR INPUT IMPAIRMENTS. SR AND CR ARE PERCENTAGES; ITER IS THE MEAN NUMBER OF ITERATIONS OVER SUCCESSFUL TRIALS. GREEN AND BLUE MARK THE BEST AND SECOND-BEST VALUES.

Method	Noisy			Laggy			Mode-Switching			Single-DoF		
	SR (↑)	CR (↓)	Iter (↓)	SR (↑)	CR (↓)	Iter (↓)	SR (↑)	CR (↓)	Iter (↓)	SR (↑)	CR (↓)	Iter (↓)
Direct	62.37	29.17	81.08	80.34	17.19	57.61	74.22	14.71	118.06	50.52	21.22	128.81
Direct + NeoSS	72.01	6.12	87.78	79.69	0.78	67.44	81.51	0.52	111.41	61.98	3.52	121.33
HO	42.84	31.64	103.02	9.37	13.41	143.65	17.97	15.23	136.56	23.05	21.61	131.24
HO + NeoSS	49.09	11.59	107.22	9.90	3.52	142.72	17.57	4.04	137.45	25.52	7.03	131.51
RT-V3	84.11	11.07	68.37	86.06	9.77	50.93	83.33	4.68	105.40	69.79	10.81	107.17
RT-V3 + NeoSS	82.16	2.34	79.78	83.98	0.26	63.05	81.12	1.95	107.53	67.58	5.60	110.41
RT-V3, unfactorized 6-D	71.22	21.88	54.38	75.91	20.96	42.24	78.64	17.50	60.98	62.50	32.68	77.63

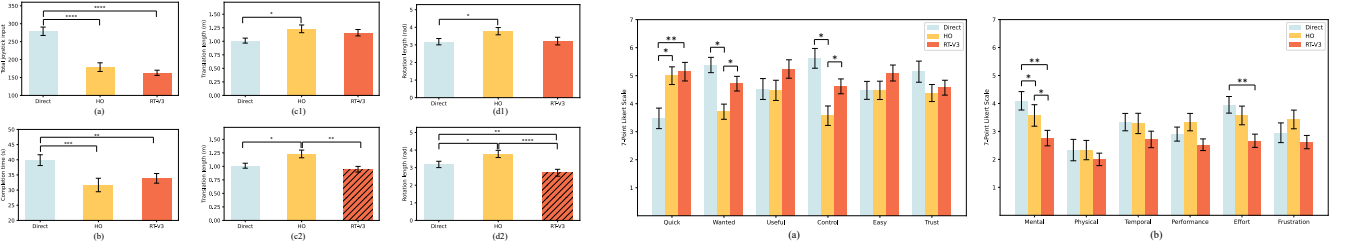


Fig. 3. Human-study results ($N = 21$). **Left:** joystick inputs (a), completion time (b), and translational (c) and rotational (d) path lengths before (1) and after (2) removing RT-V3 prior-only segments. **Right:** agreement ratings (a; higher is better) and NASA-TLX ratings (b; lower is better). * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$, and **** = $p < 0.0001$.

scene contained two or three objects, and a text instruction specified the target object and required approach direction. The interface separately exposed translation, rotation, final approach, and gripper actions. A trial failed following an unrecoverable collision or joint-limit violation, a disruptive scene change (unmatched), or a 90-s timeout. After each block, participants completed a six-item agreement survey and the NASA Task Load Index (NASA-TLX) [10]. All participants provided written consent, and the institutional ethics committee approved the protocol (G-2024-8370).

RT-V3 has the descriptively highest success rate and the lowest unmatched and timeout rates (Table II). For successful trials, Friedman tests found overall method effects on total joystick input ($p < 0.0001$) and completion time ($p = 0.0006$). Pairwise comparisons with two-stage Benjamini–Hochberg correction showed that RT-V3 required fewer joystick inputs ($p < 0.0001$) and shorter completion time ($p = 0.01$) than Direct (Fig. 3). This reduction is consistent with the asynchronous controller: once a trajectory becomes sufficiently clear, the user can pause while the robot continues the coarse approach.

Participants rated RT-V3 as enabling quicker completion than Direct ($p = 0.0078$) and gave it higher *Wanted* and *Control* ratings than HO ($p = 0.0375$ and $p = 0.0113$). Descriptively, RT-V3 also has the lowest mean NASA-TLX score on every subscale. Relative to Direct, participants reported lower mental demand and effort with RT-V3 ($p = 0.0298$ and $p = 0.0035$). Together, these results suggest that RT-V3 reduces perceived interaction burden while preserving

a stronger sense of control than HO.

Raw translational and rotational path lengths exhibit overall method effects ($p = 0.0352$ and $p = 0.0286$, respectively). After removing segments in which RT-V3 acted from its prior alone, overall effects remain for both adjusted measures ($p = 0.0009$ and $p < 0.0001$). RT-V3’s adjusted rotational path is shorter than Direct’s; for both adjusted measures, RT-V3 and Direct are shorter than HO (Fig. 3).

IV. DISCUSSION AND CONCLUSION

The experiments reveal an important gap between open-loop forecasting and interactive control. Although the unfactorized 6-D model yields the lowest prediction errors, the factorized model is substantially more reliable in closed-loop execution. Point-cloud context likewise provides a modest forecasting benefit but raises planning success from 78.8% to 90.4%, showing that prediction error alone is insufficient for selecting an interactive controller.

Remaining failures arise when planner demonstrations differ from human motion or when the candidate grasp set omits the desired affordance. The human evaluation is also limited to 21 novices performing short tabletop tasks. These limitations motivate assistance that falls back to direct control under high uncertainty or sustained user–controller conflict, together with broader evaluation across users, interfaces, and tasks. Overall, results from trajectory forecasting, a 1,000-trial planning benchmark, 768 shared-control target scenarios, and the participant study support trajectory-level probabilistic control for cluttered $SE(3)$ grasping.

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